From pattern to wave – A field-ontological interpretation of light, matter and scales (UQSH)

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Abstract

This work develops a field-ontological perspective on light, matter, and space within the framework of the Universal Quantum Foam Hypothesis (UQSH). Radiation does not appear as an object in space, but rather as the movement of the field itself, and matter as bound interference of this movement. Starting from the spherical nature of all emission, it is shown how superposition, coherence, and geometric feedback lead to the emergence of stable structures. Light, shock fronts, and cosmic waves are understood as different scale readings of

the same process. The text integrates quantum phenomena, gravity, and cosmology into a unified picture, directly continuing the UQSH, in which space is already understood as the active carrier of all physical processes. The aim is to provide a coherent framework that does not replace existing theories, but rather unifies them at a deeper ontological level.

1 Radiation as the limiting dynamics of the field

In UQSH, the different types of radiation are not separate messengers of different physical regimes, but rather different expressions of one and the same field-mechanical process. Radiation always arises where a field state loses its binding, where a local node, condensation, or vortex structure can no longer be maintained. What is called emission in classical physics is, in this view, the return of bound motion to free field motion.

Radiation is therefore not an addition to matter, but its complementary state. Matter is bound field motion, radiation is released field motion. Both belong to the same medium and differ not in their essence, but in their degree of organization. A stable pattern maintains tension in the field. If this stability breaks, the medium does not relax in a point-like manner, but rather in a wave-like pattern. This relaxation propagates through the field as a vibration, and this vibration is precisely what constitutes radiation.

The frequency of this radiation is not an external label, but is directly coupled to the internal tension of the previously bound pattern. A weakly bound state releases the field in slow, long-wavelength oscillations, such as radio waves, microwaves, or infrared. Strongly stressed, highly dense patterns release themselves in fast, short-wavelength oscillations such as ultraviolet, X-rays, or gamma rays. The frequency is thus the imprint of the previously stored field voltage. It carries the memory of the strength of order that could no longer be maintained. One can imagine the field as an elastic continuum that is locally overstretched or overtightened. As long as a pattern is stable, the medium retains this deformation. If the stability breaks down, the field does not react with an abrupt jump, but with a continuous rearrangement. This rearrangement propagates as a wave. Radiation is therefore not a movement through a medium, but a form of movement of the medium itself.

Crucially, this dynamic is scale-invariant. That we measure it as light is solely due to our coupling. At the atomic level, these field oscillations appear as photons of specific energies. At the cosmic level, they manifest as shock fronts, jets, magnetic field waves, or background radiation. In both cases, it is the same process: field tension is converted into free motion. The categories light, radiation, wave, or shock front do not denote different substances, but rather different interpretations of the same process.

This becomes particularly clear when considering the example of a stellar emission. The sun is not a point emitting particles, but a permanently oscillating field node. Every physical process within it, nuclear fusion, pressure waves, magnetic turbulence, is a local deformation of the field. This deformation propagates not in a directed manner, but isotropically. Every emission is inherently spherical. The fact that we nevertheless speak of light rays is already a perspectival reduction. Our eyes and instruments couple only to a narrow frequency range. Within this window, the spherical field response appears as a directed current because we ourselves are local patterns of the same field. We are, in a sense, cutting a line out of the sphere. The large spherical wave and the light are therefore not two phenomena, but one and the same process on different levels of description. On a fine scale, it manifests as a periodic field oscillation. On a coarse scale, it appears as a spatial stress front of the medium. As with a water wave, which appears macroscopically as a ring and microscopically as a molecular vibration, it is not a matter of two movements, but of a single dynamic with different levels of resolution.

Radiation is thus the visible transition from order to motion. Every wave is the trace of a pattern that no longer exists. It is the echo of a structure that has lost its geometric identity. In this sense, light is not merely a physical phenomenon among others, but the elementary expression of the field's ability to transform tension into motion.

2 Superposition, interference, and the formation of matter

In the context of UQSH, superposition is not a random encounter of particles, but a geometric process within the field itself. Every radiation is an ongoing field deformation, a rhythmic change of state. When two or more such deformations meet, they do not simply add up, but together form a new stress profile in the medium. The field does not calculate these movements discretely, but continuously. It forms a common pattern of gradients, phases, and curvatures from them. This can be visualized like water waves. Two intersecting waves do not create a third substance, but a temporary pattern of peaks and troughs. Briefly, zones of increased amplitude and zones of relative calm arise. This is precisely what happens in the field, but not on a surface, rather throughout the entire volume of the medium. As long as this superposition remains flat, it remains radiation. The waves penetrate each other, interfere, and separate again. The field can elastically propagate these stresses.

But at a certain point, something else happens. When several oscillations coincide in such a way that their gradients no longer cancel each other out but instead reinforce each other locally, a region arises where the field can no longer propagate the shape. The deformation becomes too steep, too dense, too coherent. The medium can no longer guide it as free motion. It is precisely there that motion tips into structure. A stable field node arises when three conditions are met simultaneously. First, the superimposed waves must be coherent, meaning they must have a stable phase relationship. Second, the interference must form a self-contained, feedback-dependent shape, such as a vortex, a ring, or a node. Third, the local gradient must exceed the elasticity of the field for free propagation. Only then does the field no longer decide for propagation but for holding. Interference becomes a pattern. Wave becomes matter.

Therefore, not every intersection of radiation is sufficient. Space is constantly full of superpositions. Most are too shallow, too disordered, or too short-lived. They dissipate again. Matter only arises where interference is spatially focused, temporally sustained, and geometrically fed back. In this sense, matter is nothing other than frozen interference, a superposition that the field can no longer release.

That multiple beams are often involved is no coincidence. Two waves can generate voltage, but they oscillate against each other. Only with a third, independent phase does a spatial depth emerge in which a vortex can close. This is why the number three appears again and again, as the minimum for stable node formation. Two generate voltage, three allow structure. Superposition is thus the normal state of the field. Matter is the rare special case in which superposition is no longer permitted. It is not the fundamental element of reality, but rather its local solidification. Every stable structure is a place where the field is forced to maintain a specific geometry. And every beam is a reminder that this geometry can also be abandoned.

3 Spherical propagation and the unity of wave and space

The biggest conceptual problem arises when we try to think of light simultaneously as a ray and as spherical propagation. In conventional physics, these two images appear as competing models: here the directed photon stream, there the spherical wavefront. In UQSH, however, they are not opposites, but two interpretations of the same field process on different levels. What we experience as a light ray is not a property of the emission itself, but a consequence of our coupling. Our eyes, instruments, and bodies are themselves stable patterns within the field. They respond only to a very narrow frequency range of this global field response. Within this window, spherical propagation appears as a directed stream because we ourselves only intersect an infinitesimal portion of the sphere. We do not experience the sphere itself, but rather the local passage of the field motion through our own structure. The large spherical wave and light are therefore not two separate phenomena. It is one and the same process, only interpreted at different levels. On a fine scale, it appears as a periodic field oscillation, as an electromagnetic wave of a specific frequency. On a coarse scale, the same process is a stress and pressure front within the medium itself, a spherical reorganization of space.

An analogy makes this clear. If you throw a stone into water, concentric rings of waves appear. If you look at the water microscopically, these rings consist of countless molecular vibrations. No one would say they are two different waves. It is the same movement, just resolved differently. It is no different in a field. The sun generates a continuous disturbance of the medium. This disturbance is simultaneously a subtle vibrational pattern, which we measure as light, and a spherical response of the field, comparable to a sound wave in space itself. The fact that we do not see both at the same time is due to our own embeddedness. We are not external observers, but rather patterns within this very field. We can only read what our structure resonates with. For a being on a different scale, the same event could appear completely different, not as a stream of photons, but as a rhythmic pulsation of all space.

Here the role of pi becomes clear. The sphere is the ideal shape that a perfectly isotropic medium assumes during a response. describes precisely this ideal, the ratio under which closed propagation occurs with minimal stress. However, no real field is perfect. It carries inhomogeneities, pre-existing conditions, and residual stresses. Therefore, every real sphere is only an approximation. It is always slightly distorted. One could say that is the geometry of the ideal, and the deviation from pi is the field's signature. This minimal uncertainty is not an error, but rather the prerequisite for structure. If the propagation were exactly spherically symmetric, there would be no difference, no direction, no possibility of coupling. Everything would dissolve into pure uniformity. Only the slight deviation, the imperfection, allows local stress gradients to arise within the spherical response. And it is precisely these gradients that we interpret as frequency, polarization, direction, and ray.

This explains how a change of state can be both global and local. At the deepest level, every excitation propagates as a spherical field response. Within this response, field inhomogeneities create subtle distortions. These distortions are what we perceive as light structure. Light is not contained within the sphere. It is the sphere, but interpreted in a non-ideal, structured form. Shock fronts, pressure waves, magnetic pulses, and electromagnetic radiation do not differ in their essence, but rather in their scale and coupling pattern. Everything is field motion. Only our perception categorizes it.

From this perspective, the distinction between space and wave loses its meaning. Space is not an empty container in which waves propagate. Space is the medium itself, vibrating. Every emission is a temporary transformation of space. Every wave is a form that space assumes to transmit tension. And every structure is a place where this transformation is no longer transmitted but held in place. Physics thus becomes the geometry of a moving medium. Light is not a message in space. It is space that moves.

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